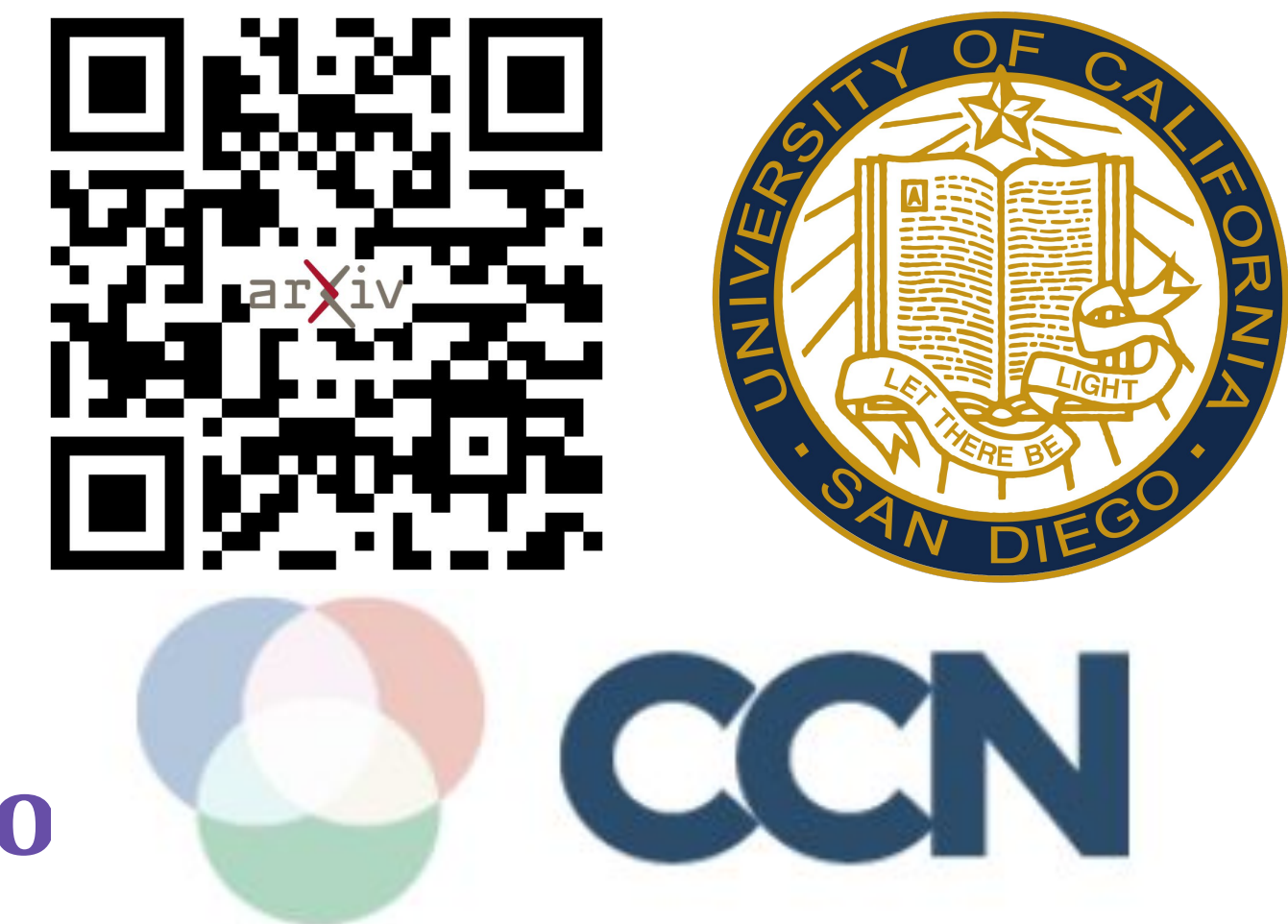


Modeling the Human Visual System: Comparative Insights from Response-Optimized and Task-Optimized Vision Models, Language Models, and Different Readout Mechanisms

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ANNs as *In Silico* Models for Brain Processing.

- Enable scalable hypothesis testing
- Reveal the 'what' and 'how' of neural information processing
- Allow causal interventions
- Allow for neural population control and BMIs

Diverse ANNs-trained on different modalities, objectives and architectures-have been used to model the visual cortex, but it's unclear which best captures each region.

We answer: What is the best way to model different regions of the visual cortex?

And propose a novel biologically grounded readout - Semantic Spatial Transformer - that sets the state of the art for predicting brain responses

Types of model inputs



Raw Visual

A woman under an umbrella standing in water on a flooded field with tents in the background.

Single Captions

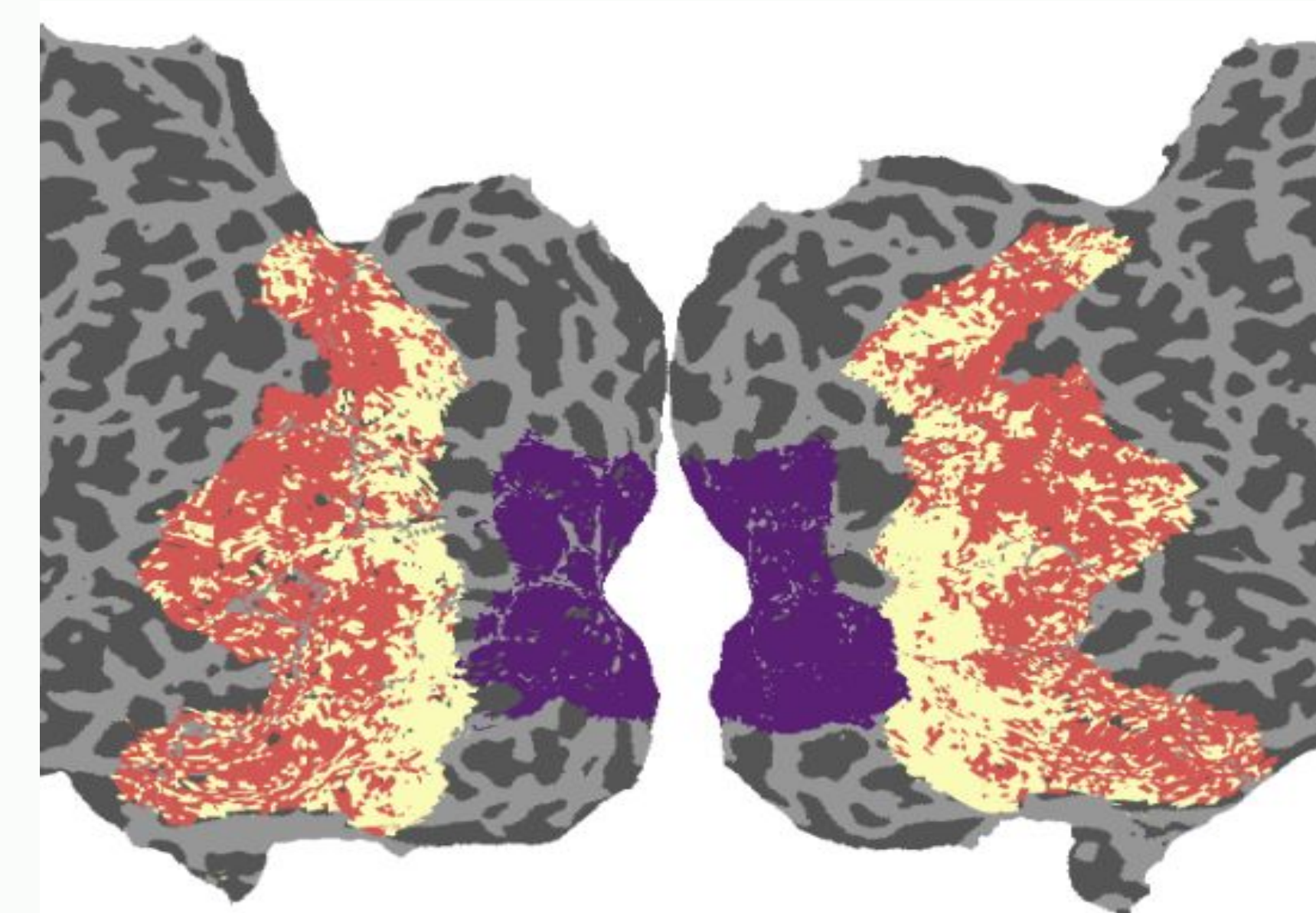
LLM Embedding



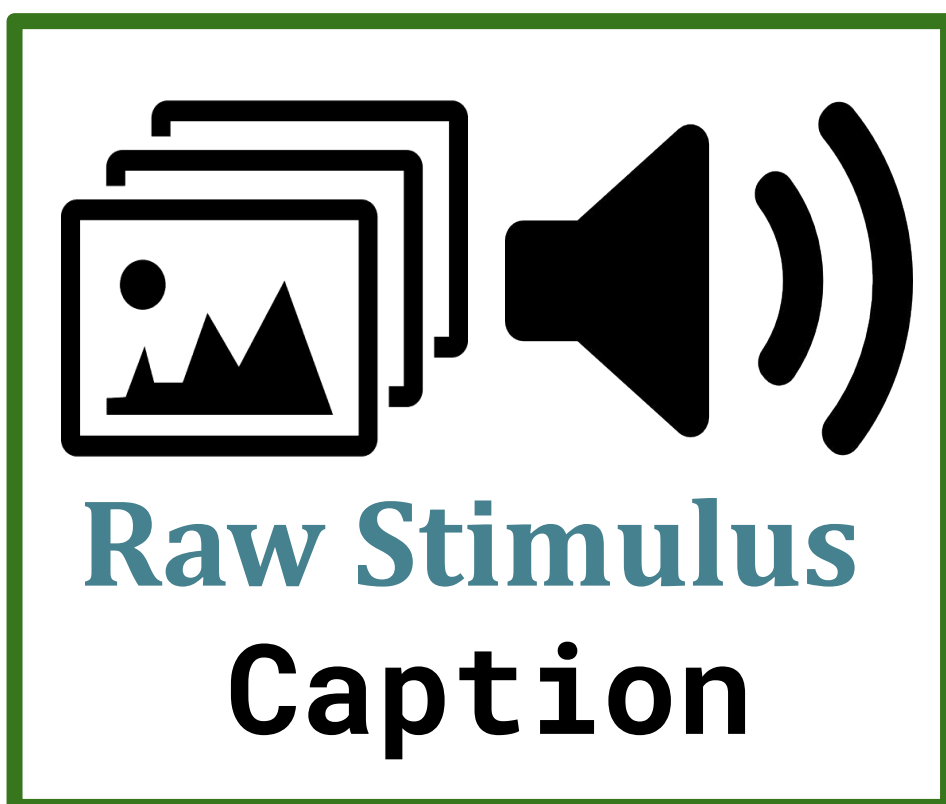
Dense Captions (Ours)

Comparing different model inputs

- Region A
- Region B
- Region C

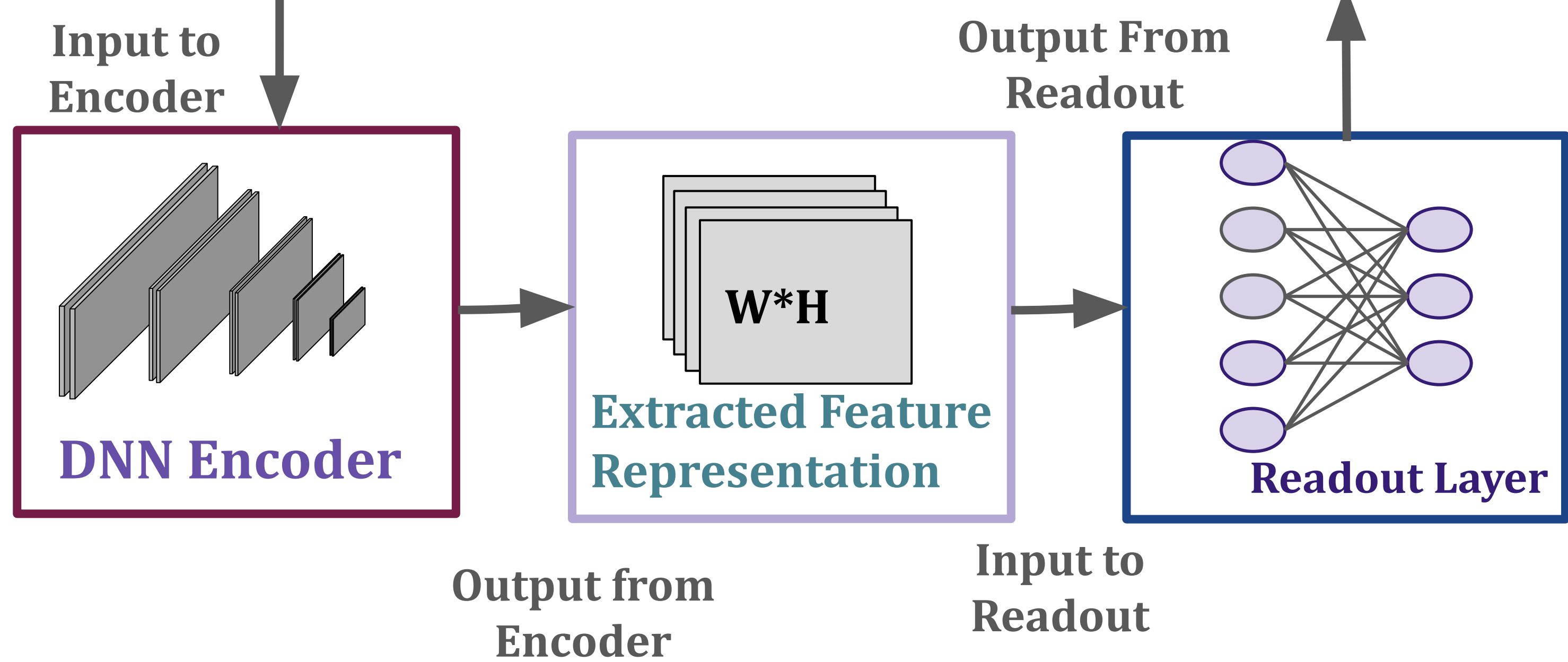


A -Low-level perceptual characteristics
B -Localized visual semantics aligned with linguistic descriptions (dense captions)
C -Global semantic interpretations aligned with language (single captions)



Raw Stimulus Caption

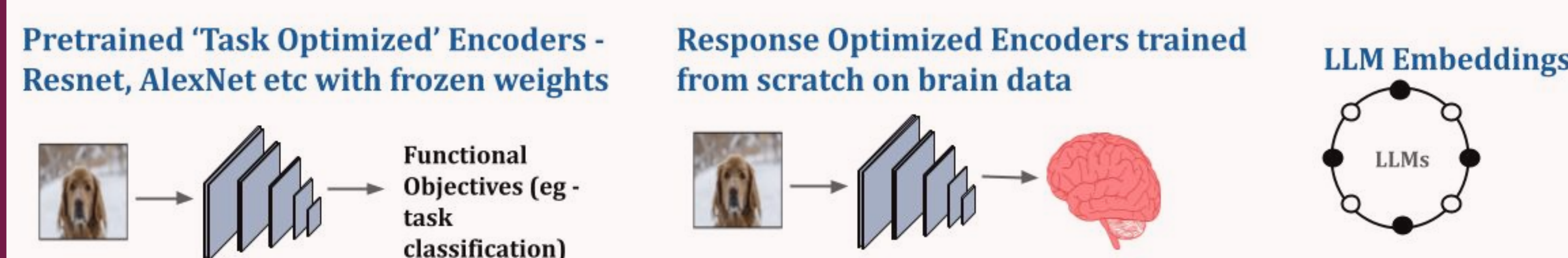
Schematic Pipeline



Output from Encoder

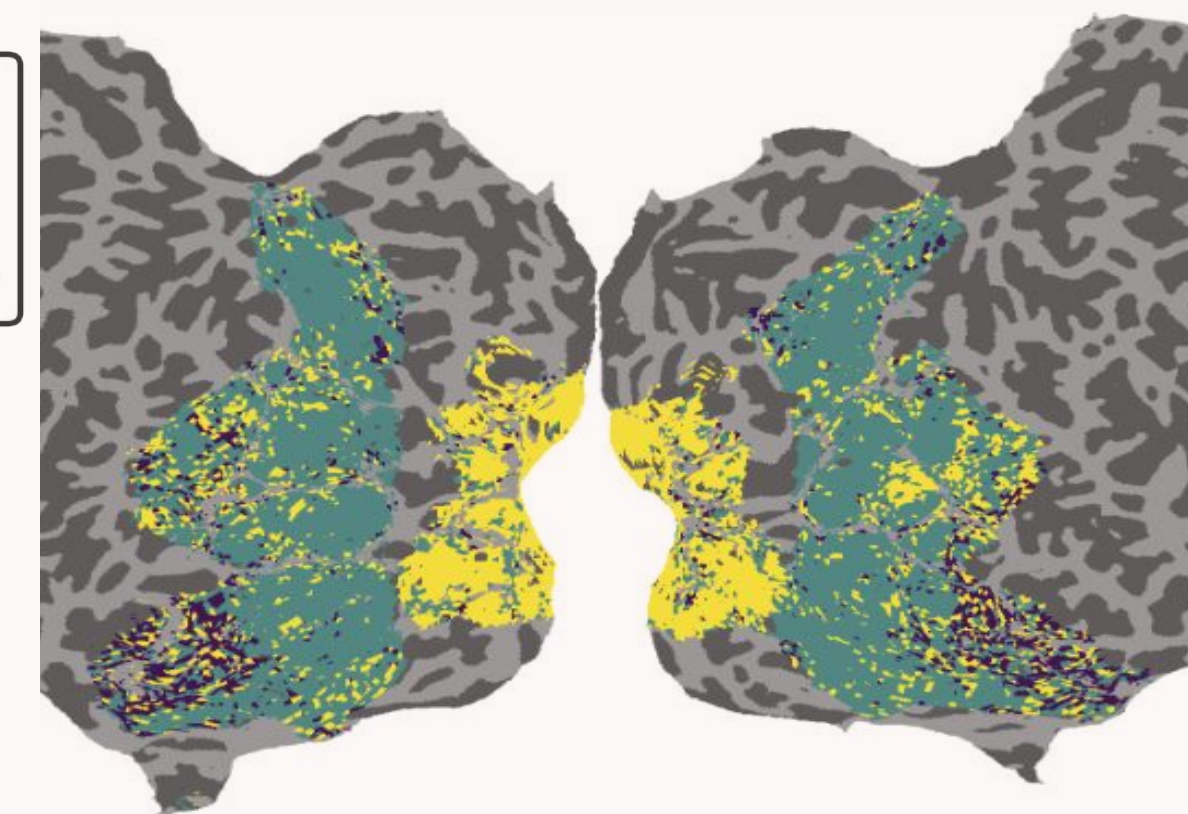
Output From Readout

Types of DNN encoders used



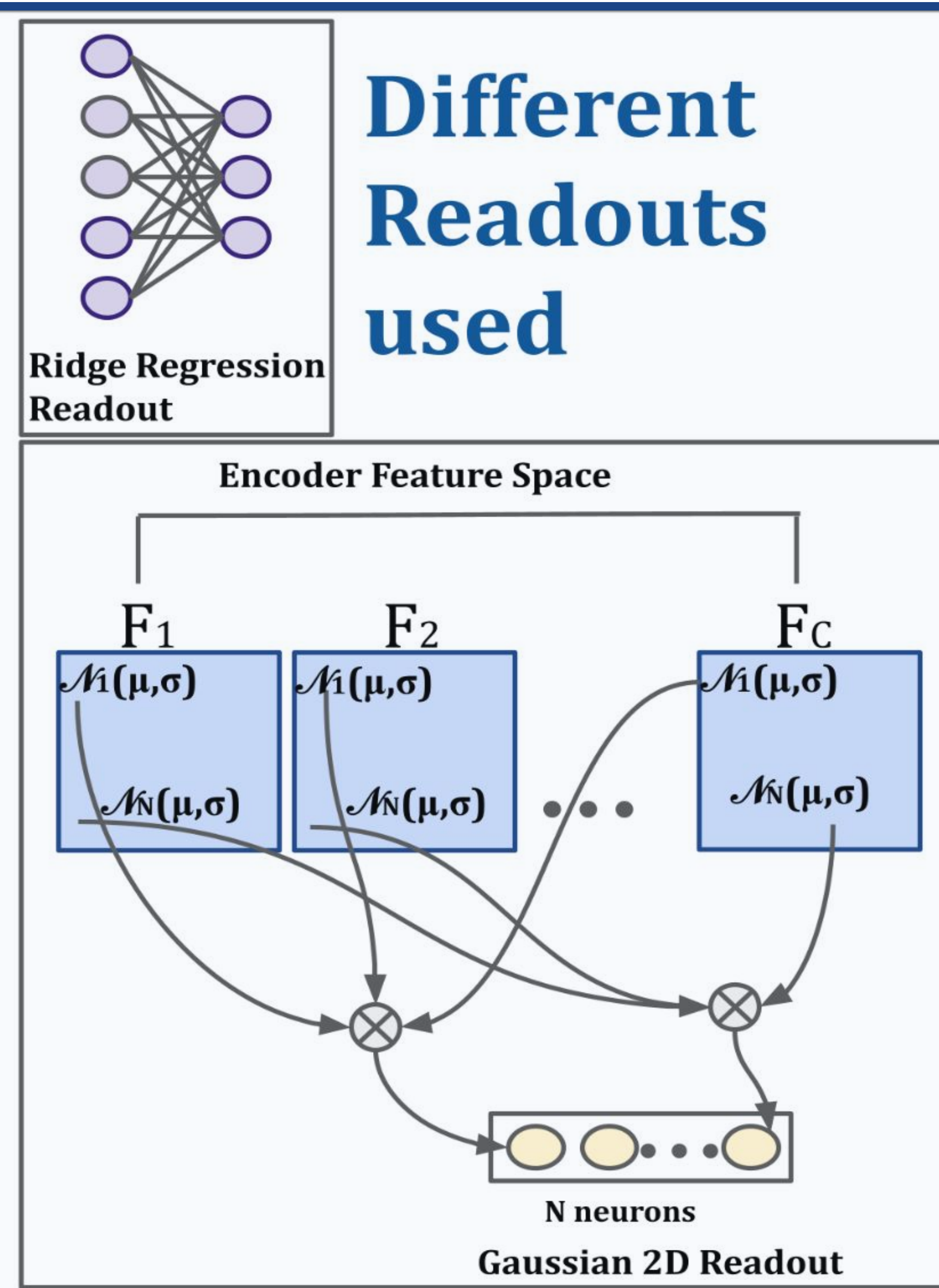
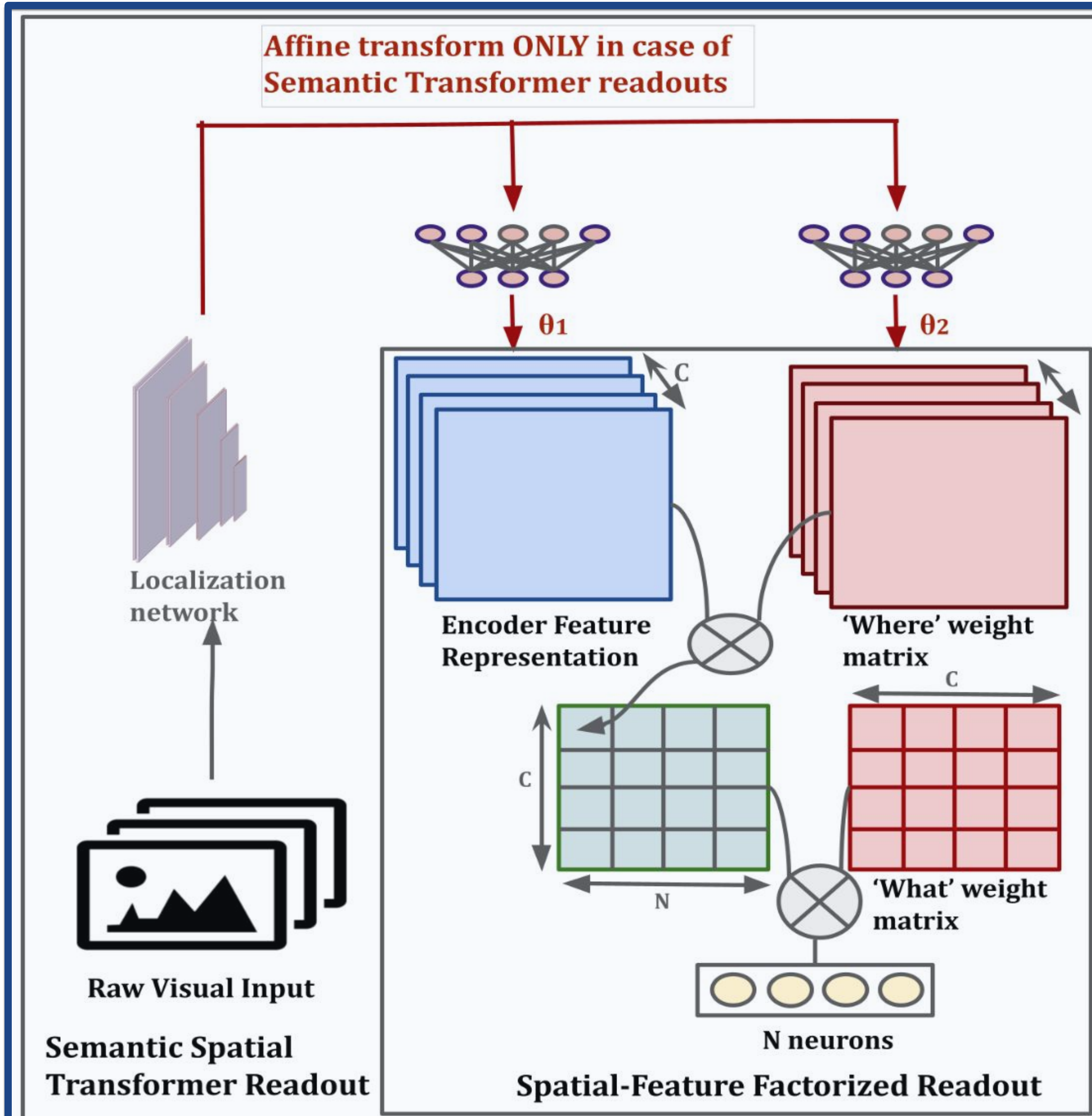
- Voxels with ≤ 0 noise ceiling
- Voxels better predicted by Task optimized encoders
- Voxels better predicted by Response optimized encoders

Optimal Encoder



References

1. Khosla, M., & Wehbe, L. (2022) - High-level visual areas act like domain-general filters with strong selectivity and functional specialization.
2. Doerig, A., Kietzmann, T. C., Allen, E., Wu, Y., Naselaris, T.Kay, K., & Charest, I. (2024). Visual representations in the human brain are aligned with large language models.
3. Klindt, D., Ecker, A., Euler, T., & Bethge, M. (2017). Neural system identification for large 579 populations separating "what" and "where."



Optimal Readout - Semantic Spatial Transformer (Ours) Readouts consistently outperformed all other existing readouts across all regions of the brain irrespective of the encoder used.

